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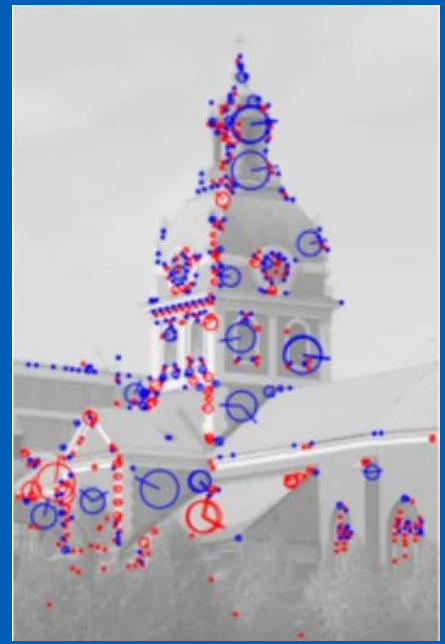
SIFT: Scale-Invariant Feature Transform



Summary

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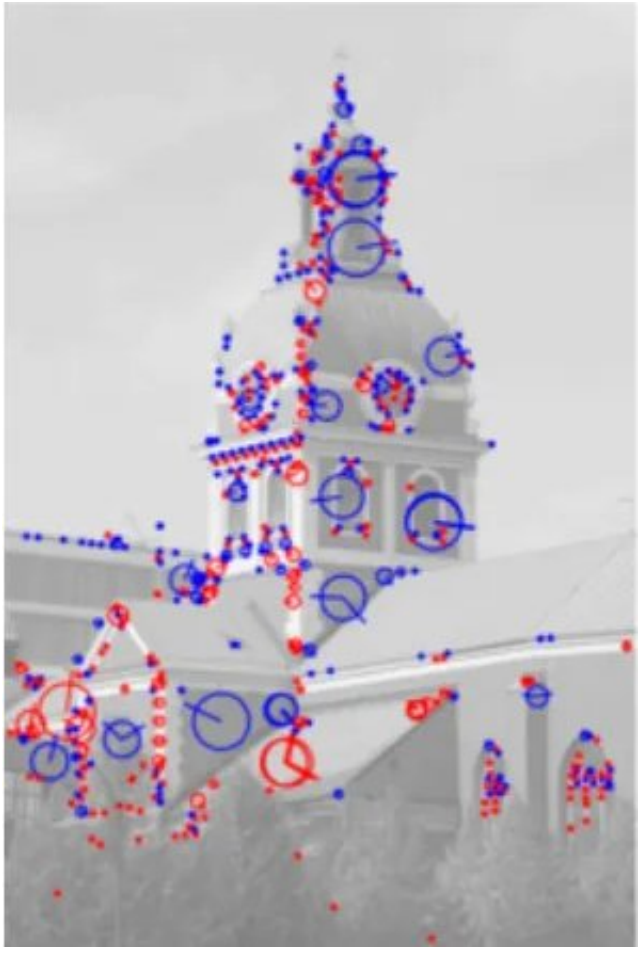
- From features to SIFT
- Stage 1 – Scale-Space Construction
- Stage 2 – Keypoint Detection
- Stage 3 – Keypoint Localization
- *Stage 4 – Orientation Assignment*
- *Stage 5 – Keypoint Descriptor*
- Invariance properties and practical notes



References

- D. G. Lowe, "Distinctive Image Features from Scale-Invariant Keypoints", IJCV 2004
  - The definitive SIFT paper
- D. G. Lowe, "Object Recognition from Local Scale-Invariant Features", ICCV 1999
  - Original SIFT conference paper
- R. Szeliski. Computer Vision: Algorithms and Applications, 2010
- CS231A – Computer Vision, Prof. S. Savarese, Stanford University
- K. Mikolajczyk, C. Schmid, "A Performance Evaluation of Local Descriptors", TPAMI 2005

Where We Are

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- Previous lecture: feature detection
  - Harris corner detector – not invariant to scale
  - Scale-invariant concept: search over window sizes / pyramid
  - SIFT mentioned as a complete scale-invariant solution
- This lecture: SIFT in depth
- SIFT provides a full pipeline
  - Detection: where are the keypoints?
  - Description: what do they look like? Next lecture!
- Goal: detect keypoints that are robust to scale, rotation, and illumination changes

SIFT: Overview

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- Proposed by David Lowe (ICCV 1999, IJCV 2004)
- Four computational stages
  - 1. Scale-Space Extrema Detection
  - 2. Keypoint Localization
  - 3. Orientation Assignment
- Invariant to scale and rotation; partially invariant to illumination

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Stage 1: Scale-Space Construction

Gaussian Scale Space

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- Goal: represent image content at multiple scales simultaneously
- Scale space: $L(x, y, \sigma) = G(x, y, \sigma) * I(x, y)$
  - G: 2D Gaussian with standard deviation σ
  - Why gaussian?
    - Lindeberg (1994): Gaussian is the only kernel that does not introduce new extrema
    - Theoretically justified choice for scale-space construction
- SIFT builds a multi-scale, multi-resolution pyramid
  - In order to build a pyramid we can independently act on (W,H) or on σ
    - (W,H) → directly modify scale
    - σ → more blurring or sharp image (coarse/fine scale)

Gaussian Scale Space

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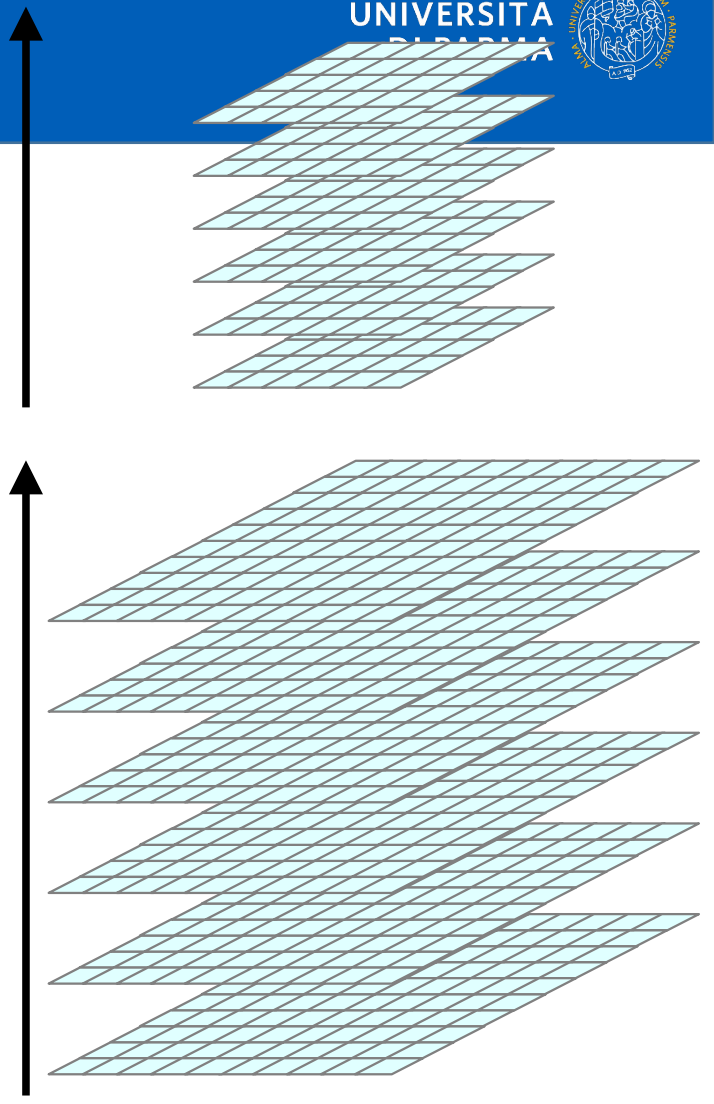
- We use octaves for $L(x, y, \sigma)$. Usual initial values
  - For each octave we have $s = 3$ levels
  - Scale factor for levels $\longrightarrow$ $k = 2^{1/s} \approx 1.26$
  - $\sigma_0 = 1.6$
  - $\sigma_i = \sigma_0 \times k^i$
- Octave 0, σ values: 1.60, 2.02, 2.54, 3.20, 4.03, 5.08 (s+3 levels, why?)
- Octave 1:
  - Downsampling by 2 starting from ~intermediate level of Octave 0 ($\sigma = 3.20$)
  - σ values: same as before, but on a different scale, namely we have to multiply by 2
  - σ actual values: 3.20, 4.03, 5.08, 6.40, 8.06, 10.16
- Ad so on…

Gaussian Scale Space

- For the Octave *o* at level *i* we have:
  $\sigma_{actual} = \sigma_0 \times k^i \times 2^o$
- How many octaves?
  - This can be computed from the original image size, usually
  - $N = \lfloor log_2(min(W,H)) \rfloor - 3$
  - Not always used, as example OpenCV has a default number (4 or 5)
- Large σ should imply large kernels → optimization trick:
  - $G(\sigma_2) = G(\sqrt{(\sigma_2^2 - \sigma_1^2)}) * G(\sigma_1)$



Laplacian of Gaussian (LoG)

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- Laplacian of Gaussian
  - $LoG(x, y, \sigma) = \nabla^2(G(x, y, \sigma) * I(x, y))$
  - Gaussian filtering ⟶ remove noise
  - Second derivative ⟶ relevant info (maybe)
    - Blobs
- Optimized as:
  - $LoG(x, y, \sigma) = \nabla^2(G(x, y, \sigma)) * I(x, y)$

Laplacian of Gaussian (LoG)

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- Blob detector: find local extrema of LoG
  - Responds maximally to blobs of a size matching σ
- Scale-normalized LoG: $\sigma^2 \cdot \nabla^2 G$
  - Without σ^2 normalization, response decreases with scale
  - Normalization needed for true scale invariance
- Extremum of $\sigma^2 \cdot \nabla^2 G$ over $(x, y, \sigma) \longrightarrow$ blob at position and scale
- Problem: LoG is expensive to compute directly

Difference of Gaussians (DoG)

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- DoG ≈ LoG (efficient approximation)
- $D(x, y, \sigma_i) = L(x, y, \sigma_{i+1}) - L(x, y, \sigma_i)$
  - DoG is the difference of two adjacent blurred images
- Relationship to LoG:
  - $D \approx (k - 1)\, \sigma^2\, \nabla^2 G$
  - $(k - 1)$ is a constant factor, does not affect extremum location
- Key advantages:
  - DoG images are computed as a by-product of building the pyramid
  - No extra filtering required

DoG Pyramid Construction

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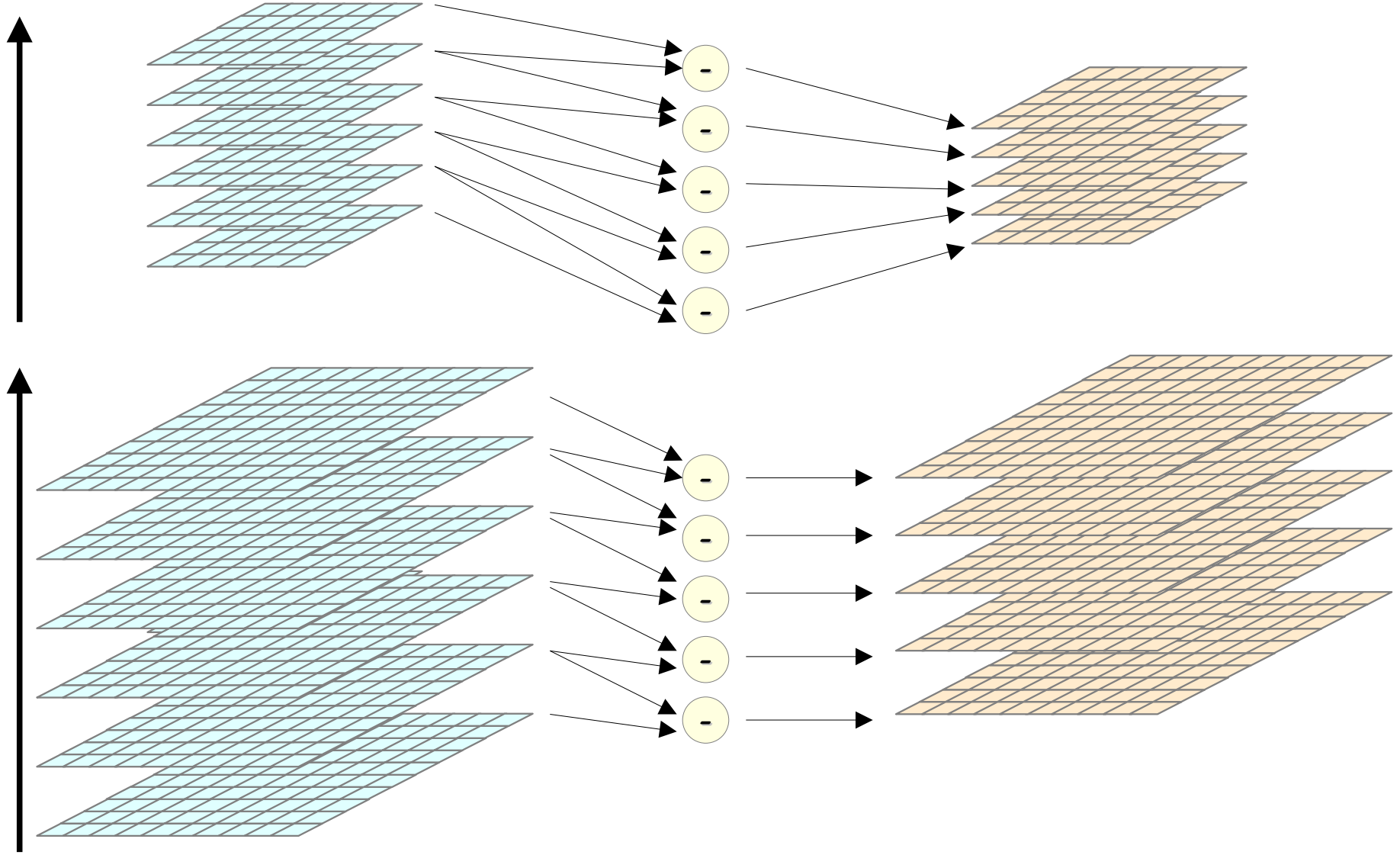
- Assuming s + 3 Gaussian images for each octave
- DoG images: adjacent Gaussian differences
  - s + 2 DoG images per octave
  - Extrema searched in the s inner DoG images
- Total keypoint search space: s levels × (octaves) × (image pixels)



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Stage 2: Keypoint Detection

Scale-Space Extrema Detection

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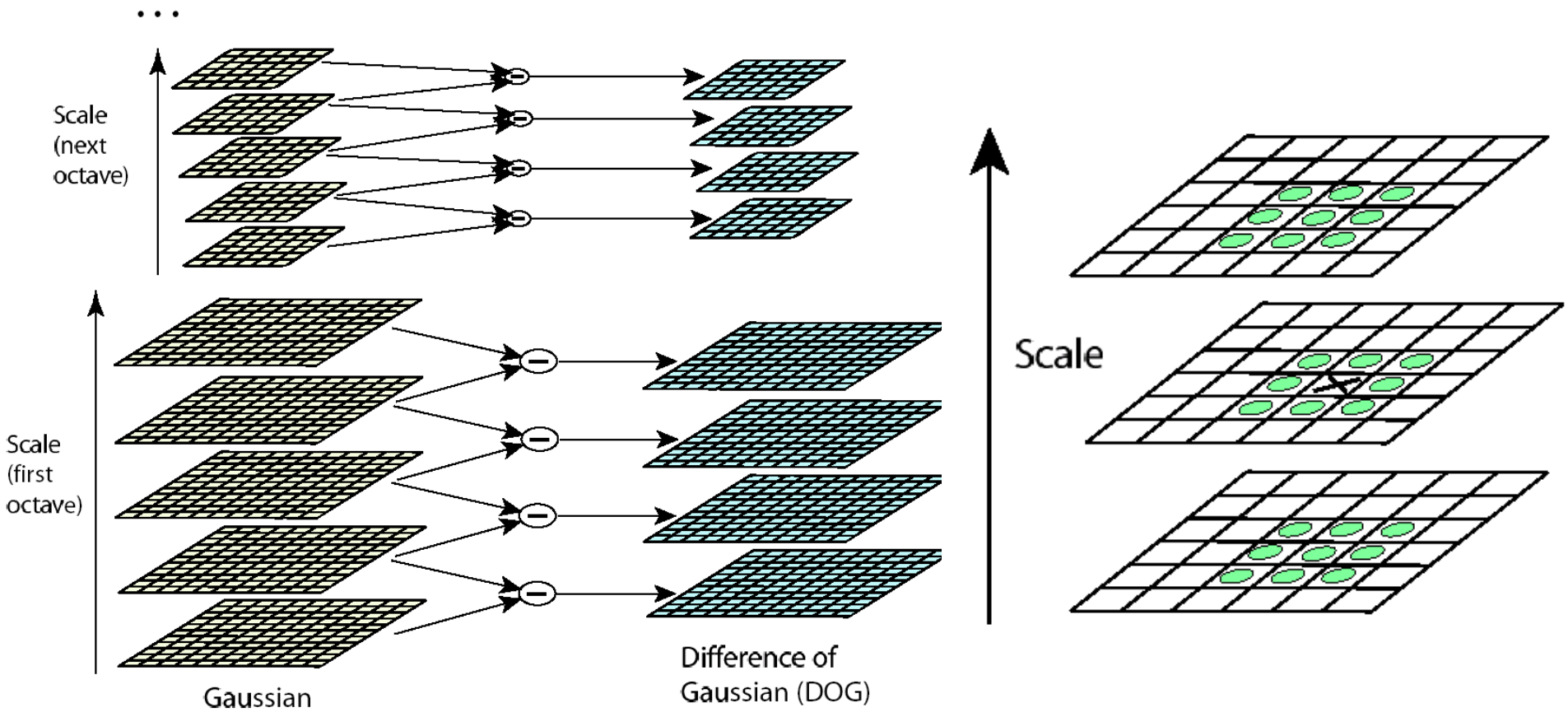
- A keypoint candidate is a local extremum of $D(x, y, \sigma_i)$
- Each sample compared to 26 neighbors
  - 8 neighbors at the same scale
  - 9 neighbors at the scale above
  - 9 neighbors at the scale below
- Selected only if larger (or smaller) than ALL 26 neighbors
- This detects scale-space blobs — local maxima and minima
- Cost
  - 3/4 of all points are eliminated after first neighbor check
  - Inexpensive: DoG is already computed

Scale-Space Extrema Detection



Voxel

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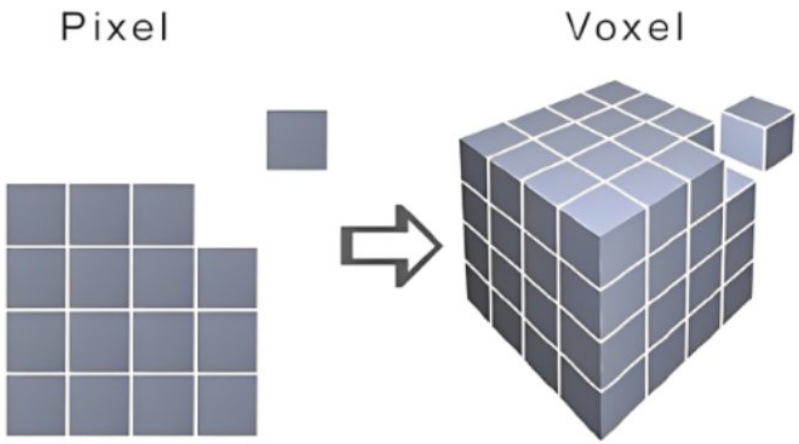
- Resulting candidates actually feature 3 coordinates
  - x, y, and σ
- They are "**voxel**s"
- The same as pixel but in a 3D grid



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Stage 3: Keypoint Localization

Accurate Localization: Taylor Expansion

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- Candidate keypoints are at discrete (pixel, scale) locations (x,y,σ)
  - SIFT goes beyond to produce sub-pixel/sub-scale interpolation
  - From (x,y,σ) to (x+Δx, y+Δy, σ+Δσ)
- D(x, y, σ) is approximated using Taylor expansion up to the second order
  - Assuming $x = (\Delta x, \Delta y, \Delta\sigma)^T$, offset from the candidate we can approximate D() as:
  - $D \approx \hat{D}(x) = D + (\partial D^T/\partial x)\, x + (1/2)\, x^T\, (\partial^2 D/\partial x^2)\, x$
- To find minima or maxima find where the derivative is 0
  - Sub-voxel offset: $\hat{x} = -(\partial^2 D/\partial x^2)^{-1}\, (\partial D/\partial x)$
- The $\hat{x}$ is the offset to be used to refine the candidate keypoints

Accurate Localization: Taylor Expansion

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- When the offset is too large we can refine the result
- If $|\hat{x}| > 0.5$ in any dimension:
  - Move the keypoint candidate to the adjacent sample
  - Repeat; typically converges in 1–3 iterations
- Improves localization accuracy
  - reduces sensitivity to sampling

Discarding Low-Contrast Keypoints

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- After Taylor fit, evaluate D at the sub-voxel location:
- $D(\hat{x}) = D + (1/2)(\partial D/\partial x)^T \hat{x}$
- Threshold check: $|D(\hat{x})| < 0.03 \longrightarrow$ discard
  - Value 0.03 is Lowe's threshold for images normalized to [0, 1]
- These points lie near zero-crossings of DoG
- They are sensitive to noise and cannot be matched reliably
- Discarding them significantly reduces the number of keypoints

Eliminating Edge Responses

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- DoG has strong responses along edges: poor localization across the edge
- 2×2 Hessian of D in image space:
  - H = [[Dxx, Dxy], [Dxy, Dyy]]
  - Eigenvalues α, β represent principal curvatures of D
- Ratio of curvatures:
  - $\mathrm{Tr}(H)^2 / \mathrm{Det}(H) = (\alpha + \beta)^2 / (\alpha\beta)$
  - For a corner: $\alpha \approx \beta \longrightarrow$ ratio close to 4
  - For an edge: $\alpha >> \beta \longrightarrow$ ratio very large
- Threshold: discard if $\mathrm{Tr}(H)^2 / \mathrm{Det}(H) > (r + 1)^2 / r$
  - Lowe uses $r = 10 \longrightarrow$ threshold = 12.1
  - Does not require computing actual eigenvalues

Keypoint Localization: Summary

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- For each DoG extremum
  - 1. Taylor expansion ⟶ sub-voxel offset $\hat{x}$
  - 2. If $|\hat{x}| > 0.5$: move to adjacent sample and repeat
  - 3. If $|D(\hat{x})| < 0.03$: discard (low contrast)
  - 4. If $Tr(H)^2/Det(H) > 12.1$: discard (edge)
- Result: stable, well-localized keypoints with assigned (x, y, σ)
- Reduction in keypoints:
  - Typically ~80–90% of initial extrema are discarded
  - Remaining keypoints are stable and distinctive

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Question time!